

ORAL AND SPOTLIGHT PRESENTATIONS

Oral & Spotlight Session 1: Augmented Reality and Video Generation Wednesday, September 24, 2025, 10:30 to 12:00 Main Hall - DCC1 Building - Room 201/202 Session Chairs: Ruogu Fang, University of Florida, United States Cheng Ouyang, University of Oxford, United Kingdom	Oral & Spotlight Session 2: Language Models - Text and Images Wednesday, September 24, 2025, 10:30 to 12:00 Dual Track Hall - DCC1 Building - Room 112 Session Chairs: Ke Zou, National University of Singapore, Singapore Keewon Shin, University of Ulsan, Asan Medical Center, Republic of Korea
Spotlight Presentations: 3D Dynamic Prediction of Missing Teeth in Diverse Patterns via Centroid-prompted Diffusion Model Authors: Zongrui Ji, Na Li, Peng Xue, Yi Dong, Lei Ma Speaker: Lei Ma, Tongji University, China Multi-expert collaboration and knowledge enhancement network for multimodal emotion recognition Authors: Kun Wang, Junyong Zhao, Liying Zhang, Qi Zhu, Daoqiang Zhang Speaker: Kun Wang, Nanjing University of Aeronautics and Astronautics, China MedGround-R1: Advancing Medical Image Grounding via Spatial-Semantic Rewarded Group Relative Policy Optimization Authors: Huihui Xu, Yuanpeng Nie, Hualiang Wang, Ying Chen, Wei Li, Junzhi Ning, Lihao Liu, Hongqiu Wang, Lei Zhu, Jiyao Liu, Xiaomeng Li, Junjun He Speaker: Huihui Xu, Shanghai AI Lab, China tHPM-LDM: Integrating Individual Historical Record with Population Memory in Latent Diffusion-based Glaucoma Forecasting Authors: Yuheng Fan, Jianyang Xie, Yimin Luo, Yanda Meng, Savita Madhusudhan, Gregory Y.H. Lip, Li Cheng, Yalin Zheng, He Zhao Speaker: Yuheng Fan, University of Liverpool, United Kingdom Lymphoma Prognosis with Lesion-Anatomy Context Fusion and Attention-Based Multi-Lesion Aggregation Authors: Song Zhang, Jiajin Zhang, Liheng Qiu, Wei Liu, Dakai Jin, Le Lu, Shenmiao Yang, Ke Yan Speaker: Ke Yan, DAMO Academy, Alibaba Group, China Occlusion-free 4D Gaussians for Open Surgery Videos Using Multi-Camera Shadowless Lamps Authors: Yuna Kato, Shohei Mori, Hideo Saito, Yoshifumi Takatsume, Hiroki Kajita, Mariko Isogawa Speaker: Mariko Isogawa, Keio University, Japan	Spotlight Presentations: DiffAtlas: GenAI-fying Atlas Segmentation via Image-Mask Diffusion Authors: Hantao Zhang, Yuhe Liu, Jiancheng Yang, Weidong Guo, Xinyuan Wang, Pascal Fua Speaker: Hantao Zhang, EPFL, Switzerland Guiding Registration with Emergent Similarity from Pre-Trained Diffusion Models Authors: Nurislam Tursynbek, Hastings Greer, Başar Demir, Marc Niethammer Speaker: Marc Niethammer, UCSD, United States Global and Local Vision-Language Alignment for Few-Shot Learning and Few-Shot OOD Detection Authors: Jie Yan, Xiaoyuan Guan, Wei-Shi Zheng, Hao Chen, Ruixuan Wang Speaker: Jie Yan, Sun Yat-sen University, China UniMRG: Refining Medical Semantic Understanding Across Modalities via LLM-Orchestrated Synergistic Evolution Authors: Hongyan Xu, Arcot Sowmya, Ian Katz, Dadong Wang Speaker: Hongyan Xu, University of New South Wales, Australia HyperPath: Knowledge-Guided Hyperbolic Semantic Hierarchy Modeling for WSI Analysis Authors: Peixiang Huang, Yanyan Huang, Weiqin Zhao, Junjun He, Lequan Yu Speaker: Peixiang Huang, The University of Hong Kong, Hong Kong SAR Towards Robust Medical Image Referring Segmentation with Incomplete Textual Prompts Authors: Qijie Wang, Xian Lin, Zengqiang Yan Speaker: Xian Lin, Huazhong University of Science and Technology, China

Oral Presentations:

Intelligent Virtual Sonographer (IVS): Enhancing Physician-Robot-Patient Communication

Authors: Tianyu Song, Feng Li, Yuan Bi, Angelos Karlas, Amir Yousefi, Daniela Branzan, Zhongliang Jiang, Ulrich Eck, Nassir Navab

Speaker: Tianyu Song, Technical University of Munich, Germany

Ophora: A Large-Scale Data-Driven Text-Guided Ophthalmic Surgical Video Generation Model

Authors: Wei Li, Ming Hu, Guoan Wang, Lihao Liu, Kaijing Zhou, Junzhi Ning, Xin Guo, Zongyuan Ge, Lixu Gu, Junjun He

Speaker: Wei Li, Shanghai Jiao Tong University, China

HieraSurg: Hierarchy-Aware Diffusion Model for Surgical Video Generation

Authors: Diego Biagini, Nassir Navab, Azade Farshad

Speaker: Diego Biagini, Technical University of Munich, Germany

Unsupervised Cardiac Video Translation Via Motion Feature Guided Diffusion Model

Authors: Swakshar Deb, Nian Wu, Frederick H. Epstein, Miaomiao Zhang

Speaker: Swakshar Deb, University of Virginia, United States

HoloPointNet: A Deep Learning Framework for Efficient 3D Point Cloud Holography

Authors: Ankit Amrutkar, Ahmet Nazlioglu, Björn Kampa, Volkmar Schulz, Johannes Stegmaier, Markus Rothermel, Dorit Merhof

Speaker: Ankit Amrutkar, RWTH Aachen University, Germany

Augmented Reality-based Guidance with Deformable Registration in Head and Neck Tumor Resection

Authors: Qingyun Yang, Fangjie Li, Jiayi Xu, Zixuan Liu, Sindhura Sridhar, Whitney Jin, Jennifer Du, Jon Heiselman, Michael Miga, Michael Topf, Jie Ying Wu

Speaker: Qingyun Yang, Vanderbilt University, United States

Oral Presentations:

Historical Report Guided Bi-modal Concurrent Learning for Pathology Report Generation

Authors: Ling Zhang, Boxiang Yun, Qingli Li, Yan Wang

Speaker: Ling Zhang, East China Normal University, China

WSI-Agents: A Collaborative Multi-Agent System for Multi-Modal Whole Slide Image Analysis

Authors: Xinheng Lyu, Yuci Liang, Wenting Chen, Meidan Ding, Jiaqi Yang, Guolin Huang, Daokun Zhang, Xiangjian He, Linlin Shen

Speaker: Xinheng Lyu, University of Nottingham Ningbo China, China

RAPTOR: Generative AI for Parsing Colorectal Cancer Referrals to Streamline Faster Diagnostic Standard Pathways

Authors: Sofiat Abioye, Shazad Ashraf, Junaid Qadir, Adam Byfield, Anusha Jose, William Poulett, Ben Wallace, Adil Butt, Colm Forde, Marcus Mottershead, Simon Fallis, Andrew Beggs, Aneel Bhangu, Lukman Akanbi, Muhammad Bilal

Speaker: Sofiat Abioye, Birmingham City University, United Kingdom

TRRG: Towards Truthful Radiology Report Generation with Cross-modal Disease Clue Enhanced Large Language Models

Authors: Yuhao Wang, Yue Sun, Tao Tan, Chao Hao, Yawen Cui, Xinqi Su, Weichen Xie, Linlin Shen, Zitong Yu

Speaker: Yuhao Wang, University of Science and Technology of China, China

A Multimodal Contrastive Learning for Detecting Aortic Dissection on 3D Non-Contrast CT with Anatomy Simplification

Authors: Duoer Zhang, Wenbo Xiao, Chen Jiang, Yuxuan Qiu, Zhan Feng, Hong Wang, Yefeng Zheng, Wentao Zhu

Speaker: Duoer Zhang, Zhejiang Lab, China

LEAVS: An LLM-based Labeler for Abdominal CT Supervision

Authors: Ricardo Bigolin Lanfredi, Yan Zhuang, Mark Finkelstein, Praveen Thoppey Srinivasan Balamuralikrishna, Luke Krembs, Brandon Khoury, Arthi Reddy, Pritam Mukherjee, Neil M. Rofsky, Ronald M. Summers

Speaker: Ricardo Bigolin Lanfredi, National Institutes of Health, United States

Oral Session 3: Image Contextualization and Multimodal Fusion Wednesday, September 24, 2025, 14:30 to 15:30 Main Hall - DCC1 Building - Room 201/202 Session Chairs: Tanveer Syeda-Mahmood, IBM Research & Stanford University, United States Yutong Xie, MBZUAI, United Arab Emirates	Oral Session 4: Detection, Segmentation, and Registration Wednesday, September 24, 2025, 14:30 to 15:30 Dual Track Hall - DCC1 Building - Room 112 Session Chairs: Johannes C. Paetzold, Cornell University, United States Zhe Xu, The Chinese University of Hong Kong, Hong Kong SAR
Multimodal Fusion Network with Distribution-based Tumor-Marker Imputation for Multi-Origin Metastatic Cervical Lymphadenopathy Classification Authors: Rui Li, Chunyan Li, Xi Lin, Jinfeng Xu, Fang Li, Yao Lu Speaker: Rui Li, Sun Yat-sen University, China A Holistic Time-Aware Classification Model for Multimodal Longitudinal Patient Data Authors: Tobias Susetzky, Huaqi Qiu, Rickmer Braren, Daniel Rueckert Speaker: Tobias Susetzky, Technical University of Munich, Germany Deep Association Multimodal Learning for Zero-shot Spatial Transcriptomics Prediction Authors: Yijing Zhou, Yadong Lu, Qingli Li, Xinxing Li, Yan Wang Speaker: Yijing Zhou, East China Normal University, China Blood Pressure Assisted Cerebral Microbleed Segmentation via Meta-matching Authors: Junmo Kwon, Jonghun Kim, Taehyeon Kim, Sang Won Seo, Hwan-ho Cho, Hyunjin Park Speaker: Junmo Kwon, Sungkyunkwan University, Republic of Korea FoundBioNet: A Foundation-Based Model for IDH Genotyping of Glioma from Multi-Parametric MRI Authors: Somayeh Farahani, Marjaneh Hejazi, Antonio Di Ieva, Sidong Liu Speaker: Somayeh Farahani, Macquarie University, Australia	Automated Characterization of Myocardial Scar Topological Patterns for Ventricular Tachycardia Screening Authors: Xicheng Sheng, Yang Zhang, Lei Li, Bailiang Chen, Freddy Odille, Xiahai Zhuang Speaker: Lei Li, National University of Singapore, Singapore Fit Pixels, Get Labels: Meta-Learned Implicit Networks for Image Segmentation Authors: Kushal Vyas, Ashok Veeraraghavan, Guha Balakrishnan Speaker: Kushal Vyas, Rice University, United States HybridMamba: A Dual-domain Mamba for 3D Medical Image Segmentation Authors: Weitong Wu, Zhaohu Xing, Jing Gong, Qin Peng, Lei Zhu Speaker: Weitong Wu, The Hong Kong University of Science and Technology (Guangzhou), China Abnormality-Driven Representation Learning for Radiology Imaging Authors: Marta Ligeró, Tim Lenz, Georg Wölflein, Omar S. M. El Nahhas, Daniel Truhn, Jakob Nikolas Kather Speaker: Marta Ligeró, EKFZ for Digital Health & TU Dresden, Germany MARSeg: Enhancing Medical Image Segmentation with MAR and Adaptive Feature Fusion Authors: Jeonghyun Hwang, Seungyeon Rhee, Minjeong Kim, Thanaporn Viriyasaranon, Jang-Hwan Choi Speaker: Jeonghyun Hwang, Ewha Womans University, Republic of Korea

Oral & Spotlight Session 5: Navigation and Surgical Workflows

Wednesday, September 24, 2025, 17:00 to 18:30
 Main Hall - DCC1 Building - Room 201/202

Session Chairs:

Yueming Jin, National University of Singapore, Singapore
 Leo Joskowicz, The Hebrew University, Israel

Spotlight Presentations:

Towards Automated Pediatric Dental Development Staging: A Dataset and Model

Authors: Peng Wang, Along He, Anli Wang, Zhenhuan Zhou, Xiaohang Guan, Tao Li

Speaker: Peng Wang, Nankai University, China

Sequence-Independent Continual Test-Time Adaptation with Mixture of Incremental Experts for Cross-Domain Segmentation

Authors: Dunnyuan Xu, Yuchen Yuan, Donghao Zhou, Xikai Yang, Jingyang Zhang, Jinpeng Li, Pheng-Ann Heng

Speaker: Dunnyuan Xu, The Chinese University of Hong Kong, Hong Kong SAR

ProgreSpine: Inherently Explainable Prototypical Regression for Spine Age Estimation

Authors: Roozbeh Bazargani, Saqib Basar, Sam Hashemi, Siavash Khallaghi

Speaker: Sam Hashemi, Prenuvo Inc, Canada

MixStyleFlow: Domain Generalization in Medical Image Segmentation using Normalizing Flows

Authors: Reza Safdari, Mohammad-Ali Nikouei Mahani, Mohamad Koohi-Moghadam, Kyongtae Tyler Bae

Speaker: Reza Safdari, The University of Hong Kong, Hong Kong SAR

Deep Learning Framework for Managing Inter-Reader Variability in Background Parenchymal Enhancement Classification for Contrast-Enhanced Mammography

Authors: Elodie Ripaud, Clément Jailin, Pablo Milioni de Carvalho, Laurence Vancamberg, Isabelle Bloch

Speaker: Elodie Ripaud, GE HealthCare, France

FedCLAM: Client Adaptive Momentum with Foreground Intensity Matching for Federated Medical Image Segmentation

Authors: Vasilis Siomos, Jonathan Passerat-Palmbach, Giacomo Tarroni

Speaker: Vasilis Siomos, City St. George's, University of London, United Kingdom

Oral & Spotlight Session 6: Towards Deployment, Bias, Trustworthiness, and Privacy

Wednesday, September 24, 2025, 17:00 to 18:30
 Dual Track Hall - DCC1 Building - Room 112

Session Chairs:

Hongwei Bran Li, National University of Singapore, Singapore
 Qingjie Meng, University of Birmingham, United Kingdom

Spotlight Presentations:

VesselVerse: A Dataset and Collaborative Framework for Vessel Annotation

Authors: Daniele Falcetta, Vincenzo Marciano, Kaiyuan Yang, Jon Cleary, Loïc Legris, Massimiliano Domenico Rizzaro, Ioannis Pitsiorlas, Hava Chaptoukaev, Benjamin Lemasson, Bjoern Menze, Maria A. Zuluaga

Speaker: Daniele Falcetta, EURECOM, France

Query-Level Alignment for End-to-End Lesion Detection with Human Gaze

Authors: Yan Kong, Zhixiang Peng, Yuan Yin, Yonghao Li, Jiangdong Cai, Sheng Wang, Qian Wang, Yuqi Fang, Caifeng Shan

Speaker: Yuqi Fang, Nanjing University, China

Bias and Generalizability of Foundation Models across Datasets in Breast Mammography

Authors: Elodie Germani, Ilayda Selin-Türk, Fatima Zeineddine, Charbel Mourad, Shadi Albarqouni

Speaker: Shadi Albarqouni, University of Bonn, Germany

Training-free Test-time Improvement for Explainable Medical Image Classification

Authors: Hangzhou He, Jiachen Tang, Lei Zhu, Kaiwen Li, Yanye Lu

Speaker: Hangzhou He, Peking University, China

A flexible deep learning framework for survival analysis with medical data

Authors: Gabriele Campanella, Ida Häggström, Lucas Kook, Torsten Hothorn, Thomas J. Fuchs

Speaker: Ida Häggström, Chalmers University of Technology, Sweden

MELON: Multimodal Mixture-of-Experts with Spectral-Temporal Fusion for Long-Term Mobility Estimation in Critical Care

Authors: Jiaqing Zhang, Miguel Contreras, Jessica Sena, Andrea Davidson, Yuanfang Ren, Ziyuan Guan, Tezcan Ozrazgat-Baslanti, Tyler J. Loftus, Subhash Nerella, Azra Bihorac, Parisa Rashidi

Speaker: Subhash Nerella, University of Florida, United States

Oral Presentations:

Towards Markerless Intraoperative Tracking of Deformable Spine Tissue

Authors: Connor Daly, Elettra Marconi, Marco Riva, Jinendra Ekanayake, Daniel S. Elson, Ferdinando Rodriguez y Baena
Speaker: Connor Daly, Imperial College London, United Kingdom

Multi-modal Representations for Fine-grained Multi-label Critical View of Safety Recognition

Authors: Britty Baby, Vinkle Srivastav, Pooja P. Jain, Kun Yuan, Pietro Mascagni, Nicolas Padoy
Speaker: Britty Baby, University of Strasbourg, France

MedAgentSim: Self-Evolving Multi-Agent Simulations for Realistic Clinical Interactions

Authors: Mohammad Almansoori, Komal Kumar, Hisham Cholakkal
Speaker: Mohammad Almansoori, Mohamed bin Zayed University of Artificial Intelligence, United Arab Emirates

EndoMamba: An Efficient Foundation Model for Endoscopic Videos via Hierarchical Pre-training

Authors: Qingyao Tian, Huai Liao, Xinyan Huang, Bingyu Yang, Dongdong Lei, Sebastien Ourselin, Hongbin Liu
Speaker: Qingyao Tian, Institute of Automation, Chinese Academy of Sciences, China

Half-SAM: SAM-based Haustral Fold Detection In Colonoscopy with Debris Suppression and Temporal Consistency

Authors: Mayank Golhar, Luojie Huang, Nicholas J. Durr
Speaker: Mayank Golhar, Johns Hopkins University, United States

SAMSA: Segment Anything Model enhanced with Spectral Angles for Hyperspectral Interactive Medical Image Segmentation

Authors: Alfie Roddan, Tobias Czempel, Chi Xu, Daniel S. Elson, Stamatia Giannarou
Speaker: Alfie Roddan, Imperial College London, United Kingdom

Oral Presentations:

Maverick: Collaboration-free Federated Unlearning for Medical Privacy

Authors: Win Kent Ong, Chee Seng Chan
Speaker: Win Kent Ong, Universiti Malaya, Malaysia

Variational Visible Layers: A Practical Framework for Uncertainty Estimation

Authors: Zeinab Abboud, Herve Lombaert, Samuel Kadoury
Speaker: Zeinab Abboud, Polytechnique Montreal, Canada

On the Interplay of Human-AI Alignment, Fairness, and Performance Trade-offs in Medical Imaging

Authors: Haozhe Luo, Ziyu Zhou, Shelley Zixin Shu, Aurélie Pahud de Mortanges, Robert Berke, Mauricio Reyes
Speaker: Haozhe Luo, University of Bern, Switzerland

Mutual Information Regularization for Fairness-aware Deep Imaging Representations

Authors: Amir Reza Sadri, Thomas DeSilvio, Satish E. Viswanath
Speaker: Amir Reza Sadri, Case Western Reserve University, United States

Adaptive Frame Selection for Gestational Age Estimation from Blind Sweep Fetal Ultrasound Videos

Authors: Tanya Akumu, Marawan Elbatel, Victor M. Campello, Richard Osuala, Carlos Martin-Isla, Ignacio Valenzuela, Xiaomeng Li, Bishesh Khanal, Karim Lekadir
Speaker: Tanya Akumu, University of Barcelona, Spain

Toward Medical Deepfake Detection: A Comprehensive Dataset and Novel Method

Authors: Shuaibo Li, Zhaoxu Xing, Hongqiu Wang, Pengfei Hao, Xingyu Li, Zekai Liu, Lei Zhu
Speaker: Shuaibo Li, The Hong Kong University of Science and Technology (Guangzhou), China

Oral Session 7: Data Quality, Restoration, and Super-resolution

Thursday, September 25, 2025, 8:30 to 9:30
 Main Hall - DCC1 Building - Room 201/202

Session Chairs:

Charles Delahunt, Global Health Labs, United States
 Kyung Sung, University of California, Los Angeles (UCLA), United States

Directional Adaptive Shuffle-Based Visual State-Space Models for Medical Image Restoration

Authors: Simon C. K. Chan, Lulin Shi, Bingxin Huang, Terence T. W. Wong

Speaker: Simon C. K. Chan, The Hong Kong University of Science and Technology, Hong Kong SAR

Advancing Medical Representation Learning Through High-Quality Data

Authors: Negin Baghbanzadeh, Adibvafa Fallahpour, Yasaman Parhizkar, Franklin Ogidi, Shuvendu Roy, Sajad Ashkezari, Vahid Reza Khazaie, Michael Colacci, Ali Etemad, Arash Afkanpour, Elham Dolatabadi

Speaker: Negin Baghbanzadeh, York University, Canada

BayeSMM: Robust Deep Combined Computing Tackling Heavy-tailed Distribution in Medical Images

Authors: Yuanye Liu, Ruoxuan Zhen, Shangqi Gao, Xinzhe Luo, Xin Gao, Qingchao Chen, Xiahai Zhuang

Speaker: Yuanye Liu, Fudan University, China

TESLA: Test-time Reference-free Through-plane Super-resolution for Multi-contrast Brain MRI

Authors: Yoonseok Choi, Sunyoung Jung, Mohammed A. Al-masni, Ming-Hsuan Yang, Dong-Hyun Kim

Speaker: Yoonseok Choi, Yonsei University, Republic of Korea

Robust Deep Learning for Myocardial Scar Segmentation in Cardiac MRI with Noisy Labels

Authors: Aida Moafi, Danial Moafi, Evgeny M. Mirkes, Gerry P. McCann, Abbas S. Alatrany, Jayanth Ranjit Arnold, Mostafa Mehdipour Ghazi

Speaker: Aida Moafi, University of Leicester, United Kingdom

Oral Session 8: Explainability and Interpretability

Thursday, September 25, 2025, 8:30 to 9:30
 Dual Track Hall - DCC1 Building - Room 112

Session Chairs:

Hao Chen, Hong Kong University of Science and Technology, Hong Kong SAR
 Zulqarnain Gilani, Edith Cowan University, Australia

An Anatomical Significance-Aware Architecture for Explainable Myocardial Infarction Prediction via Multi-Task Learning

Authors: Jiachuan Peng, Marcel Beetz, Abhirup Banerjee, Min Chen, Vicente Grau

Speaker: Jiachuan Peng, University of Oxford, United Kingdom

Uncertainty-Supervised Interpretable and Robust Evidential Segmentation

Authors: Yuzhu Li, An Sui, Fuping Wu, Xiahai Zhuang

Speaker: An Sui, Fudan University, China

Automated Detection of BK Virus in H&E Whole-Slide Images Using Weakly-Supervised Deep Learning and Interpretable Morphological Biomarkers

Authors: Sharifa Sahai, Ana D. Ramos-Guerra, Cristina Almagro-Pérez, Guillaume Jaume, Andrew Zhang, Helmut Rennke, Astrid Weins, Juan E. Ortuño, Maria J. Ledesma-Carbayo, Faisal Mahmood

Speakers: Sharifa Sahai, Harvard Medical School and Ana D. Ramos-Guerra, Universidad Politécnica de Madrid, Spain

X-SiT: Inherently Interpretable Surface Vision Transformers for Dementia Diagnosis

Authors: Fabian Bongratz, Tom Nuno Wolf, Jaume Gual Ramon, Christian Wachinger

Speaker: Fabian Bongratz, Technical University of Munich, Germany

Latent Motion Profiling for Annotation-free Cardiac Phase Detection in Adult and Fetal Echocardiography Videos

Authors: Yingyu Yang, Qianye Yang, Kangning Cui, Can Peng, Elena D'Alberty, Netzahualcoyotl Hernandez-Cruz, Olga Patey, Aris T. Papageorghiou, J. Alison Noble

Speaker: Yingyu Yang, University of Oxford, United Kingdom

Oral Session 9: Generalizability and Adaptation Thursday, September 25, 2025, 14:30 to 15:30 Main Hall - DCC1 Building - Room 201/202 Session Chairs: Peirong Liu, Johns Hopkins University, United States Guotai Wang, University of Electronic Science and Technology of China, China	Oral Session 10: Multimodal and Longitudinal Prediction Thursday, September 25, 2025, 14:30 to 15:30 Dual Track Hall - DCC1 Building - Room 112 Session Chairs: Jang-Hwan Choi, Ewha Womans University, Republic of Korea Chenyu You, Stony Brook University, United States
tCURLoRA: Tensor CUR Decomposition Based Low-Rank Parameter Adaptation and Its Application in Medical Image Segmentation Authors: Guanghai He, Wangang Cheng, Hancan Zhu, Xiaohao Cai, Gaohang Yu Speaker: Guanghai He, Shaoxing University, China	Multi-scale Attention-based Multiple Instance Learning for Breast Cancer Diagnosis Authors: Mariana Mourão, Jacinto C. Nascimento, Carlos Santiago, Margarida Silveira Speaker: Mariana Mourão, Institute for Systems and Robotics, Instituto Superior Técnico, Portugal
Reverse Imaging for Wide-spectrum Generalization of Cardiac MRI Segmentation Authors: Yidong Zhao, Peter Kellman, Hui Xue, Tongyun Yang, Yi Zhang, Yuchi Han, Orlando Simonetti, Qian Tao Speaker: Yidong Zhao, Delft University of Technology, the Netherlands	Parameterized Diffusion Optimization enabled Autoregressive Ordinal Regression for Diabetic Retinopathy Grading Authors: Qinkai Yu, Wei Zhou, Hantao Liu, Yanyu Xu, Meng Wang, Yitian Zhao, Huazhu Fu, Xujiang Ye, Yalin Zheng, Yanda Meng Speaker: Qinkai Yu, University of Exeter, United Kingdom
Frequency Strikes Back: Boosting Parameter-Efficient Foundation Model Adaptation for Medical Imaging Authors: Son T. Ly, Hien V. Nguyen Speaker: Son T. Ly, University of Houston, United States	Attention-Based Multimodal Deep Learning Model for Post-Stroke Motor Impairment Prediction Authors: Rukiye Karakis, Kali Gurkahraman, Georgios D. Mitsis, Marie-Hélène Boudrias Speaker: Rukiye Karakis, Sivas Cumhuriyet University, Turkey
Tied Prototype Model for Few-Shot Medical Image Segmentation Authors: Hyeongji Kim, Stine Hansen, Michael Kampffmeyer Speaker: Hyeongji Kim, The Arctic University of Norway, Norway	DISCLOSE the Neurodegeneration Dynamics: Individualized ODE Discovery for Alzheimer's Disease Precision Medicine Authors: Wooseok Jung, Joonhyuk Park, Won Hwa Kim Speaker: Joonhyuk Park, Pohang University of Science and Technology, Republic of Korea
Sparsely Labeled fMRI Data Denoising with Meta-Learning-Based Semi-Supervised Domain Adaptation Authors: Keun-Soo Heo, Ji-Wung Han, Soyeon Bak, Minjoo Lim, Bogyong Kang, Sang-Jun Park, Weili Lin, Han Zhang, Dinggang Shen, Tae-Eui Kam Speaker: Keun-Soo Heo, Korea University, Republic of Korea	From Pixels to Prognosis: A Multi-Modal Attention-based Framework for Visceral Adipose Tissue Estimation Authors: Arooba Maqsood, Afsah Saleem, Marc Sim, David Suter, Simone Radavelli-Bagatini, Jonathan M. Hodgson, Richard L. Prince, Kun Zhu, William D. Leslie, John T. Schousboe, Joshua R. Lewis, Syed Zulqarnain Gilani Speaker: Arooba Maqsood, Edith Cowan University, Australia

Oral & Spotlight Session 11: Graphs - Learning and Data

Thursday, September 25, 2025, 17:00 to 18:30
 Main Hall - DCC1 Building - Room 201/202

Session Chairs:

Angelica Aviles-Rivero, Tsinghua University, China
 Jelmer Wolterink, University of Twente, the Netherlands

Spotlight Presentations:

PathVG: A New Benchmark and Dataset for Pathology Visual Grounding

Authors: Chunlin Zhong, Shuang Hao, Junhua Wu, Xiaona Chang, Jiwei Jiang, Xiu Nie, He Tang, Xiang Bai

Speaker: Shuang Hao, Huazhong University of Science and Technology, China

Endoscopic Artifact Inpainting for Improved Endoscopic Image Segmentation

Authors: Zhangyuan Yu, Chenlin Du, Hongrui Liang, Xiuqi Zheng, Zeyao Ma, Mingjun Wu, Mingwu Ao, Qicheng Lao

Speaker: Qicheng Lao, Beijing University of Posts and Telecommunications, China

DEFUSE-MS: Deformation Field-Guided Spatiotemporal Graph-Based Framework for Multiple Sclerosis New Lesion Detection

Authors: Mostafa Salem, Salma Hassan, Vijay Ram Kumar Papineni, Ayman Elsayed, Mohammad Yaqub

Speaker: Mostafa Salem, Mohamed Bin Zayed University of Artificial Intelligence, United Arab Emirates

Bipartite Patient-Modality Graph Learning with Event-Conditional Modelling of Censoring for Cancer Survival Prediction

Authors: Hailin Yue, Hulin Kuang, Jin Liu, Junjian Li, Lanlan Wang, Mengshen He, Jianxin Wang

Speaker: Hailin Yue, Central South University, China

Stronger Together: Registering Preoperative Imagery, LUS, and MIS Liver Images

Authors: Mohammad Mahdi Kalantari, Erol Ozgur, Mohammad Alkhatib, Navid Rabbani, Yamid Espinel, Richard Modrzejewski, Bertrand Le Roy, Emmanuel Buc, Youcef Mezouar, Adrien Bartoli

Speaker: Mohammad Mahdi Kalantari, Clermont-Auvergne INP, France

A Novel Streamline-based diffusion MRI Tractography Registration Method with Probabilistic Keypoint Detection

Authors: Junyi Wang, Mubai Du, Ye Wu, Yijie Li, William M. Wells III, Lauren J. O'Donnell, Fan Zhang

Speaker: Junyi Wang, UESTC, China

Oral & Spotlight Session 12: Reconstruction and Synthesis

Thursday, September 25, 2025, 17:00 to 18:30
 Dual Track Hall - DCC1 Building - Room 112

Session Chairs:

Cosmin I. Bercea, Technical University of Munich, Germany
 Chuyang Ye, Beijing Institute of Technology, China

Spotlight Presentations:

Cross-view Generalized Diffusion Model for Sparse-view CT Reconstruction

Authors: Jixiang Chen, Yiqun Lin, Yi Qin, Hualiang Wang, Xiaomeng Li

Speaker: Jixiang Chen, The Hong Kong University of Science and Technology, Hong Kong SAR

Hallucination-Aware Multimodal Benchmark for Gastrointestinal Image Analysis with Large Vision-Language Models

Authors: Bidur Khanal, Sandesh Pokhrel, Sanjay Bhandari, Ramesh Rana, Nikesh Shrestha, Ram B. Gurung, Cristian Linte, Angus Watson, Yash R. Shrestha, Binod Bhattarai

Speaker: Binod Bhattarai, University of Aberdeen, United Kingdom

Veriserum: A dual-plane fluoroscopic dataset with knee implant phantoms for deep learning in medical imaging

Authors: Jinhao Wang, Florian Vogl, Pascal Schütz, Saša Čuković, William R. Taylor

Speaker: Jinhao Wang, ETH Zürich, Switzerland

Unisyn: A Generative Foundation Model for Universal Medical Image Synthesis across MRI, CT and PET

Authors: Yulin Wang, Honglin Xiong, Kaicong Sun, Jiameng Liu, Xin Lin, Ziyi Chen, Yanzhe He, Qian Wang, Dinggang Shen

Speaker: Yulin Wang, ShanghaiTech University, China

Inter-class separability loss for weakly supervised mutually exclusive multiclass segmentation of brain tumor lesions

Authors: Vivek Dhamale, Vaanathi Sundaresan

Speaker: Vivek Dhamale, Indian Institute of Science, Bangalore, India

Facial Appearance Prediction with Conditional Multi-scale Autoregressive Modeling for Orthognathic Surgical Planning

Authors: Jungwook Lee, Xuanang Xu, Daeseung Kim, Tianshu Kuang, Hannah H. Deng, Xinrui Song, Yasmine Soubra, Rohan Dharia, Michael A.K. Liebschner, Jaime Gateno, Pingkun Yan

Speaker: Jungwook Lee, Rensselaer Polytechnic Institute, United States

Oral Presentations:

Cerebrovascular Diseases Screening from Color Fundus Photography via Cross-View Fusion and Graph-Based Discrimination

Authors: Congyu Tian, Shihao Zou, Xiangyun Liao, Cheng Chen, Chubin Ou, Jianping Lv, Shanshan Wang, Weixin Si
Speaker: Shihao Zou, Shenzhen Institutes of Advanced Technology, China

Multi-Modal Graph-Based Machine Learning for Predicting Surgical Outcome in Epilepsy Patients

Authors: Artur Arturi Aharonyan, Syeda Abeera Amir, Nunthasiri Wittayanakorn, Marius George Linguraru, Chima Oluigbo, Syed Muhammad Anwar
Speaker: Artur Arturi Aharonyan, Childrens National Hospital, United States

SG2VID: Scene Graphs Enable Fine-Grained Control for Video Synthesis

Authors: Ssharvien Kumar Sivakumar, Yannik Frisch, Ghazal Ghazaei, Anirban Mukhopadhyay
Speaker: Ssharvien Kumar Sivakumar, TU Darmstadt, Germany

GLCP: Global-to-Local Connectivity Preservation for Tubular Structure Segmentation

Authors: Feixiang Zhou, Zhuangzhi Gao, He Zhao, Jianyang Xie, Yanda Meng, Yitian Zhao, Gregory Y.H. Lip, Yalin Zheng
Speaker: Feixiang Zhou, University of Liverpool, United Kingdom

pyOpenNFT: an Open-Source Python Framework for ML-based Real-time fMRI and EEG-fMRI Neurofeedback

Authors: Ekaterina Antipushina, Nikita Davydov, Riccardo De Feo, Evgeny Prilepin, Artem Nikonorov, Yury Koush
Speaker: Ekaterina Antipushina, Skolkovo Institute of Science and Technology, Russia

GPU Accelerated Modeling of Cortical Radial and Tangential Connectivity Changes in Neurodegeneration

Authors: Hongbo Zhang, Xinyu Nie, Jiaxin Yue, Yuan Li, John Ringman, Yonggang Shi
Speaker: Hongbo Zhang, University of Southern California, United States

Oral Presentations:

VesselGPT: Autoregressive Modeling of Vascular Geometry

Authors: Paula Feldman, Martin Sinnona, Claudio Delrieux, Viviana Siless, Emmanuel Iarussi
Speaker: Paula Feldman, Universidad Torcuato Di Tella, Argentina

ClipGS: Clippable Gaussian Splatting for Interactive Cinematic Visualization of Volumetric Medical Data

Authors: Chengkun Li, Yuqi Tong, Kai Chen, Zhenya Yang, Ruiyang Li, Shi Qiu, Jason Ying-Kuen Chan, Pheng-Ann Heng, Qi Dou
Speaker: Chengkun Li, The Chinese University of Hong Kong, Hong Kong SAR

InstructX2X: An Interpretable Local Editing Model for Counterfactual Medical Image Generation

Authors: Hyungi Min, Taeseung You, Hangyeul Lee, Yeongjae Cho, Sungzoon Cho
Speaker: Hyungi Min, Seoul National University, Republic of Korea

UFO-3: Unsupervised Three-Compartment Learning for Fiber Orientation Distribution Function Estimation

Authors: Xueqing Gao, Rizhong Lin, Jianhui Feng, Yonggang Shi, Yuchuan Qiao
Speaker: Xueqing Gao, Tongji University, China

3D Acetabular Surface Reconstruction from 2D Pre-operative X-ray Images using SRVF Elastic Registration and Deformation Graph

Authors: Shuai Zhang, Jinliang Wang, Xu Wang, Sujith Konan, Danail Stoyanov, Evangelos B. Mazomenos
Speaker: Shuai Zhang, University College London, United Kingdom

Speech Audio Generation from Dynamic MRI via a Knowledge Enhanced Conditional Variational Autoencoder

Authors: Yaxuan Li, Han Jiang, Yifei Ma, Shihua Qin, Jonghye Woo, Fangxu Xing
Speaker: Yaxuan Li, The University of Hong Kong, Hong Kong SAR

Oral Session 13: Surgical Data Science

Friday, September 26, 2025, 8:30 to 9:30
 Main Hall - DCC1 Building - Room 201/202

Session Chairs:

Vinkle Srivastav, IHU Strasbourg, University of
 Strasbourg, France
 Mobarak I. Hoque, University of Manchester, United
 Kingdom

Endo-FASt3r: Endoscopic Foundation Model Adaptation for Structure from Motion

Authors: Mona Sheikh Zeinoddin, Mobarak I. Hoque, Zafer
 Tandogdu, Greg L. Shaw, Matthew J. Clarkson, Evangelos B.
 Mazomenos, Danail Stoyanov

Speaker: Mona Sheikh Zeinoddin, UCL Hawkes Institute,
 University College London, United Kingdom

RadGS-Reg: Registering Spine CT with Biplanar X-rays via Joint 3D Radiative Gaussians Reconstruction and 3D/3D Registration

Authors: Ao Shen, Xueming Fu, Junfeng Jiang, Qiang Zeng, Ye
 Tang, Zhengming Chen, Luming Nong, Feng Wang, S. Kevin Zhou

Speaker: Ao Shen, Hohai University, China

Memory-Augmented SAM2 for Training-Free Surgical Video Segmentation

Authors: Ming Yin, Fu Wang, Xujiang Ye, Yanda Meng, Zeyu Fu

Speaker: Ming Yin, University of Exeter, United Kingdom

Endo3R: Unified Online Reconstruction from Dynamic Monocular Endoscopic Video

Authors: Jiaxin Guo, Wenzhen Dong, Tianyu Huang, Hao Ding,
 Ziyi Wang, Haomin Kuang, Qi Dou, Yun-Hui Liu

Speaker: Jiaxin Guo, The Chinese University of Hong Kong, Hong
 Kong SAR

Mamba Guided Boundary Prior Matters: A New Perspective for Generalized Polyp Segmentation

Authors: Tapas K. Dutta, Snehashis Majhi, Deepak Ranjan Nayak,
 Debesh Jha

Speaker: Deepak Ranjan Nayak, Malaviya National Institute of
 Technology Jaipur, India

Oral Session 14: Missing Data and Weak Supervision

Friday, September 26, 2025, 8:30 to 9:30
 Dual Track Hall - DCC1 Building - Room 112

Session Chairs:

Cheng Chen, The University of Hong Kong, Hong
 Kong SAR
 Dmitry Dylov, The Skolkovo Institute of Science and
 Technology, Russia

ReCo-I2P: An Incomplete Supervised Lymph Node Segmentation Framework Based on Orthogonal Partial- instance Annotation

Authors: Litingyu Wang, Ping Ye, Wenjun Liao, Shichuan Zhang,
 Shaoting Zhang, Guotai Wang

Speaker: Litingyu Wang, University of Electronic Science and
 Technology of China, China

Tree-based Semantic Losses: Application to Sparsely- supervised Large Multi-class Hyperspectral Segmentation

Authors: Junwen Wang, Oscar Maccormac, William Rochford,
 Aaron Kujawa, Jonathan Shapey, Tom Vercauteren

Speaker: Junwen Wang, King's College London, United Kingdom

Towards Globally Predictable k-Space Interpolation: A White-box Transformer Approach

Authors: Chen Luo, Qiyu Jin, Taofeng Xie, Xuemei Wang, Huayu
 Wang, Congcong Liu, Liming Tang, Guoqing Chen, Zhuo-Xu Cui,
 Dong Liang

Speaker: Chen Luo, Inner Mongolia University, China

Anatomical Graph-based Multilevel Distillation for Robust Alzheimer's Disease Diagnosis with Missing Modalities

Authors: Fei Liu, Huabin Wang, Mohamed Hisham Jaward,
 Shiuan-Ni Liang, Huey Fang Ong, Jiayuan Cheng

Speaker: Fei Liu, Monash University Malaysia, Malaysia

DC-Seg: Disentangled Contrastive Learning for Brain Tumor Segmentation with Missing Modalities

Authors: Haitao Li, Ziyu Li, Yiheng Mao, Zhengyao Ding,
 Zhengxing Huang

Speaker: Haitao Li, Zhejiang University, China